

HealthConnect Clinic

Project Summary

Executive Summary

The Week 4 Data Analytics work for HealthConnect Clinic establishes the foundation for investigating appointment attendance and no-show behaviour. The focus was to use appointment level data to understand which measurable characteristics may be associated with missed appointments and to define a clear framework for deeper analysis. This stage prioritises data understanding and analytical planning rather than premature conclusions.

Project Objective

The Data Analytics track addresses the question: what patterns in the available appointment data are associated with attendance outcomes, particularly no-shows, and which measures should be monitored to support future operational decisions? Week 4 therefore establishes the variables, questions and KPIs required to answer this question in later analysis.

Work Completed in Week 4

The dataset and Data Dictionary were reviewed; appointment_id was identified as the primary appointment identifier; duplicate appointment IDs were checked with no duplicates identified; limited missing values were identified in distance_to_clinic_km and waiting_time_minutes; relevant analytical variables were documented; business questions and proposed KPIs were defined; and the risks, limitations and next-stage analytical approach were documented.

Data Quality and Integrity Findings

The primary key check produced a positive result: no duplicate appointment_id values were identified. These supports using appointment_id as the appointment-level identifier for subsequent analysis. The Data Dictionary also indicates that patient_id can legitimately occur across multiple appointments, so repeated patient IDs should not automatically be treated as duplicate records.

Although distance_to_clinic_km and waiting_time_minutes each contain approximately 1% empty values. The level of missingness is limited, these fields require deliberate treatment before detailed analysis. Missing distance or waiting time should not automatically be replaced with zero because zero has a substantive meaning and would create potentially misleading records.

Key Considerations

Key considerations are missing values, data quality, correct interpretation of appointment outcomes, the possibility that important attendance factors are not captured, and the need to avoid treating associations as proof of causation.

Week 5 Priorities

The immediate priority for Week 5 is to move from planning into evidence generation. The analysis should begin with a baseline description of appointment outcomes, followed by KPI calculations and segmented comparisons. Attention should be paid to `previous_no_shows`, `reminder_sent`, `booking_lead_days`, `appointment_day`, `appointment_time` and `appointment_type` because these variables have strong analytical relevance to the no-show problem.

The output of the next stage should be a concise set of defensible findings rather than a long list of observations. Each finding should state what the data shows, identify the relevant segment or measure, explain why it matters to HealthConnect and indicate what should be investigated or considered next.

Conclusion

Week 4 has established a structured foundation for the HealthConnect Data Analytics track. The dataset is sufficiently rich to support meaningful investigation of appointment outcomes, and the primary appointment identifier has passed the duplicate check. Limited missingness has been identified in two numerical variables and will require deliberate treatment. The selected variables, business questions and KPI framework now provide a clear route into deeper analysis. The next stage should focus on quantifying the no-show problem, identifying the strongest patterns and converting those patterns into evidence-based business insights.